Integrating Multiresolution and Multitemporal Sentinel-2 Imagery for Land-Cover Mapping in the Xiongan New Area, China

Xin Luo, *Member, IEEE*, Xiaohua Tong, *Senior Member, IEEE*, and Haiyan Pan

***Abstract*— Accurate land use/land cover (LULC) mapping over a large area is essential to environmentally sustainable development. Recently, the Chinese government established a new national economic zone called the Xiongan New Area, and along with the upcoming large-scale urban construction, this area will inevitably experience a dramatic LULC change, which will threaten the local ecological balance. In this article, we proposed a two-stage approach for LULC mapping in the Xiongan New Area ahead of the forthcoming dense urban construction. The first stage is to obtain base-class maps through a supervised imagery classification. Specifically, we designed a new object-based framework consisting of automatic image segmentation, pixel-based probabilistic estimation, and area-weighted probability statistics for Sentinel-2 multiresolution imagery classification. The second stage is an LULC map refinement process in which the temporal features of each land use category are extracted to refine the LULC classification. Through the implementation of the proposed two-stage approach, an LULC map containing permanent water, temporal water, natural vegetation, barren land, built-up land, and cropland categories can be produced. Through an accuracy assessment, the proposed multiresolution imagery classification method achieved the highest overall accuracy of 88.58% and an average accuracy (AA) of 87.78% compared with conventional classification methods. After obtaining the refined LULC map, we find that the current Xiongan New Area is in a less developed state, that is, cropland accounts for the highest proportion of 51.59%, which is followed by natural vegetation (22.38%) and built-up land (15.69%).**

***Index Terms*— Land-cover classification, multiresolution imagery, object-based image analysis (OBIA), Sentinel-2 imagery, temporal features.**

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I. Introduction

LAND use/land cover (LULC) dynamics reflect the underlying natural and social processes in urban growth, thus providing essential information for understanding their influence on the structure and function of ecosystems [1]–[4]. On April 1, 2017, the Chinese government established a national economic zone called the Xiongan New Area in the northern province of Hebei. Launching of this new area is expected to build a world-class, green, modern, and smart city. With the forthcoming acceleration of large-scale urbanization, intensive anthropogenic activities would be imposed upon the land surface and thus lead to a land use/land cover change (LULCC) from natural vegetation or agricultural land, to a low-density built environment, then to a high-density built environment, and finally to perpetual sprawl of major urban centers [5]–[7]. To achieve sustainable urban development, timely and accurate LULC monitoring is of great importance for a plethora of applications in the Xiongan New Area, such as in decision-making concerning issues related to land management and planning, ecosystem monitoring, and protection [8]–[11].

Remote sensing has the capacity to reveal dynamic spatial patterns of LULC over a large geographic area in a consistent way; therefore, it has long been used for the LULC classification and change analyses [12]–[17]. In general, the satellite-data-based LULC classification typically relies on the distinguishing features of land-cover types in imagery, particularly, spectral features, which have been extensively exploited for imagery classification [18], [19]. However, despite some promising developments, this method is an ongoing and challenging research topic primarily because using spectral features alone sometimes falls short in detecting land cover over large geographic regions or highly spatially heterogeneous environments due to "ill-defined" problems, such as high within-class variability of individual materials, complex multimodal class compositions, and high interclass similarities [20]–[22].

Previous efforts have focused on spatial information combination for LULC classification enhancement, and it is common to use hand-engineered spatial features, such as the morphological attribute [23] and rotation invariant spectral–spatial feature representation [24]. By treating

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extracted/selected spatial features as an additional input to supervised classification, the spatial information can be combined to generate an improved classification map. Object-based image analysis (OBIA) uses a wealth of spectral and spatial information to enable more precise LULC mapping [25]. Rather than classifying individual pixels through only spectral features, object-based classification first groups spatially connected pixels into homogeneous "objects," and the objects obtained from the image segmentation analysis may, in principle, form a more suitable spatial unit than pixels for land-cover mapping because they should relate to natural spatial units. Accordingly, a direct classification can be carried out using object features, such as shape, textures, and spatial relations, by taking the objects as the basic unit [26]–[28]. The other way to obtain object-based classification results is to use a majority voting (MV) strategy within each object based on pixel-wise classification [29], [30]. Moreover, Markov random fields (MRFs) model the contextual correlation among image pixels in terms of conditional prior probabilities of individual pixels given their neighboring pixels; thus, the spatial dependences within a pixel neighborhood can also be used to support the prediction of the central pixel [31], [32].

The combined use of spatial information can achieve further improvement in classification performance, particularly at high spatial resolutions [33], [34]. The Earth's surface varies over time with specific inherent mechanisms as reported by Chen *et al.* [35], and the related temporal features with respect to land-cover types can also be depicted to further improve the classification performance. Many studies have demonstrated the usefulness of capturing phenology-significant differences in vegetation cover classification [12], [36]–[39]. Moreover, landscape change caused by climate, ecosystems, hydrology, and human activities, among other factors, also involves time-related land-cover change principles, such as the expansion of building surfaces with time and the shrinking of wetland in the dry season. All these principles can be used to facilitate the LULC classification.

In addition to the classification method, a high-precision LULC classification also depends on the accessibility and availability of remote sensing data (e.g., multispectral images, hyperspectral images, and images with high, moderate, or low spatial resolution). Very recently, the European Space Agency (ESA) launched the Sentinel-2A and Sentinel-2B satellites in 2015 and 2017, respectively. Sentinel-2 imagery has many advantages, such as high spatial resolution (up to 10 m), high revisitation (five days), wide swath width (290 km), many band settings (13 spectral bands), and free data accessibility. Due to the superior properties in terms of spectral, spatial, and temporal resolutions, Sentinel-2 imagery is particularly applicable for continuous LULC monitoring over large regions. However, new Sentinel-2 imagery presents high specificity, and very few existing methods can fully combine information that covers the spectral, spatial, and temporal domains for LULC mapping. In this article, we selected Sentinel-2 images to map the LULC of the Xiongan New Area. To fully exploit the spectral, spatial, and temporal information of the imagery, we proposed a new object-based LULC classification framework. Thus, three main contributions of this article were, as follows.

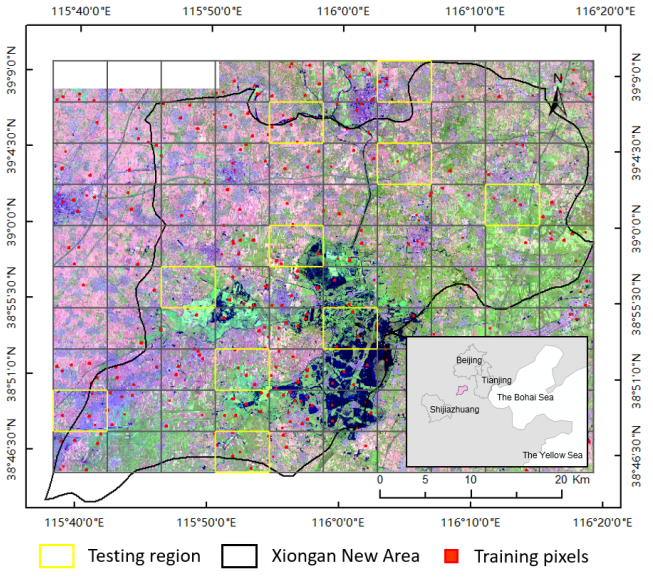


Fig. 1. Study area and validation regions (the location of the Xiongan New Area is given by the State Bureau of Surveying and Mapping). The Sentinel-2 imagery acquired on September 30, 2016, is displayed as a false color composite of the bands at the central wavelengths of 1610, 865, and 560 nm.

1) A new LULC classification framework tailored for the multiresolution and multitemporal Sentinel-2 imagery is proposed. This new framework integrates the techniques of image classification, image segmentation, object-based area weighting, and temporal feature characterization and can fully exploit spectral, spatial, and temporal information of the image for the LULC classification.
2) To support the OBIA, we adopt a split-merge strategy and propose a fully automatic image segmentation method.
3) Ahead of the large-scale urban development in the Xiongan New Area, a high-precision LULC thematic map, which is of great value in land resource management, environmental conservation, and urban expansion monitoring, has been produced.

The remainder of this article is structured as follows. In Section II, we introduce the study area and the related remote sensing data. In Section III, the details of the two-stage approach are introduced, and in Section IV, the related results are presented, including the LULC map and an accuracy evaluation. In Sections V and VI, an in-depth discussion and conclusions associated with the methodology and results are provided, respectively.

II. Data Sets

A. Study Area

The Xiongan New Area, which is approximately 2000 km^2, consists of the three underdeveloped counties of Xiongxian, Rongcheng, and Anxin in eastern China (see Fig. 1). This area includes the largest freshwater wetland in North China, that is, Baiyangdian. Due to the benefits of the Baiyangdian wetland, this area maintains both a scenic natural landscape and high eco-environmental quality. Because the economic zone of the Xiongan New Area is planned, this area is regarded as another

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TABLE I
MULTITEMPORAL AND MULTIRESOLUTION SENTINEL-2 IMAGERY USED IN THE LULC CLASSIFICATION STUDY

Test imagery	Multiresolution imagery			Multitemporal imagery	
	Band	Resolution	Wavelength range (nm)	Acquisition date	Overall landscape characteristics
Sentinel-2 MSI imagery	2 Blue	10 m	458-523	July 22, 2016	Large-scale exposed soil at sowing time Decreased surface water in the drying season
	3 Green		543-578		
	4 Red		650-680	Aug. 31, 2016	Large-scale green vegetation at the ripening time; Surface water shrinkage in the drying season
	8 Near infrared		785-900		
	5 Red edge 1	20 m	698-713	Sept. 30, 2016	Large-scale exposed soil at the harvest time;
	6 Red edge 2		733-748		
	7 Red edge 3		773-793		
	8a Near infrared narrow		855-875		
	11 Shortwave infrared 1		1565-1655	Feb. 27, 2017	Water stored in the Baiyangdian wetland in wintertime
	12 Shortwave infrared 2		2100-2280		

new area of national significance after the Shenzhen Special Economic Zone and the Shanghai Pudong New Area in China. With the goal of becoming a first-class international city, the construction of the Xiongan New Area is planned to cover approximately 100 km^2 in the initial stage and further expand to 2000 km^2 in the long term. Therefore, in the foreseeable future, the Xiongan New Area would be changed dramatically from a variety of aspects due to the rapid urban construction, particularly in the aspect of LULC transition.

B. Data Collection

More recently, through the European Earth Observation program Copernicus, the Sentinel-2 satellite pair, which aims at data continuity for global land surface monitoring along with other satellites, was launched on June 23, 2015, and March 7, 2017. Unlike most satellite imagery, which has only one finer resolution panchromatic band, the Sentinel-2 multispectral instrument (MSI) is designed to acquire multiresolution, multispectral imagery. It involves four spectral bands at a spatial resolution of 10 m, six spectral bands at a spatial resolution of 20 m, and three spectral bands at a spatial resolution of 60 m. Because the 60-m imagery is mainly dedicated to atmospheric corrections and cloud screening [42], in this article, we select the 10-m imagery and 20-m imagery for the LULC classification. Moreover, to refine the LULC classification using temporal features, multitemporal Sentinel-2 images corresponding to different acquisition dates are collected in our study. All the Sentinel-2 imagery is downloaded from the Sentinels Scientific Data Hub (https://scihub.copernicus.eu/) as a Level-1C geometrically corrected top-of-atmosphere (TOA) reflectance product. Detailed descriptions of the selected Sentinel-2 imagery are provided in Table I.

III. METHODOLOGY

Atmospheric correction and image registration are first carried out for the acquired image in our study. Then, a two-stage approach tailored for the Sentinel-2 imagery is implemented. Specifically, the first stage aims to produce a high-precision base-class map based on the multiresolution imagery. It consists of three modules: 1) automatic split-merge image segmentation based on Sentinel-2 finer resolution; 2) pixel-based support vector machine (SVM) probabilistic classification of Sentinel-2 multiresolution images; and 3) an area-weighted average method for land-cover category determination. The second stage is an LULC map refinement process, performed based on multitemporal images, in which the temporal features corresponding to the LULC categories are extracted and applied for LULC map refinement. Finally, an accuracy assessment is carried out accordingly.

A. Preprocessing

All data acquired by the Sentinel-2 MSI have been systematically processed into TOA Level-1C orthoimage products by the Payload Data Ground Segment (PDGS). In addition, we conduct an atmospheric correction for bottom-of-atmosphere (BOA) corrected reflectance product generation using Sen2cor module version 2.2.1 within the Sentinel-2 Toolbox (S2TBX) in the Sentinel Application Platform (SNAP) version 4.0.2. In addition, image-to-image coregistration among multitemporal imageries is undertaken. We manually select at least 20 control points from the 10-m imagery followed by second-order polynomial model implementation, through which all multitemporal images are rectified with root-mean-square errors (RMSEs) of less than 1 m. Because multispectral registration for any pair of spectral bands has already been conducted with a standard deviation of 0.1 pixels [42], no such corrections are applied in the preprocessing. Moreover, through artificial image interpretation based on the Sentinel-2 10-m imagery and Google Earth high-resolution imagery, spectrally homogeneous patches distributed randomly within the study area are collected as the

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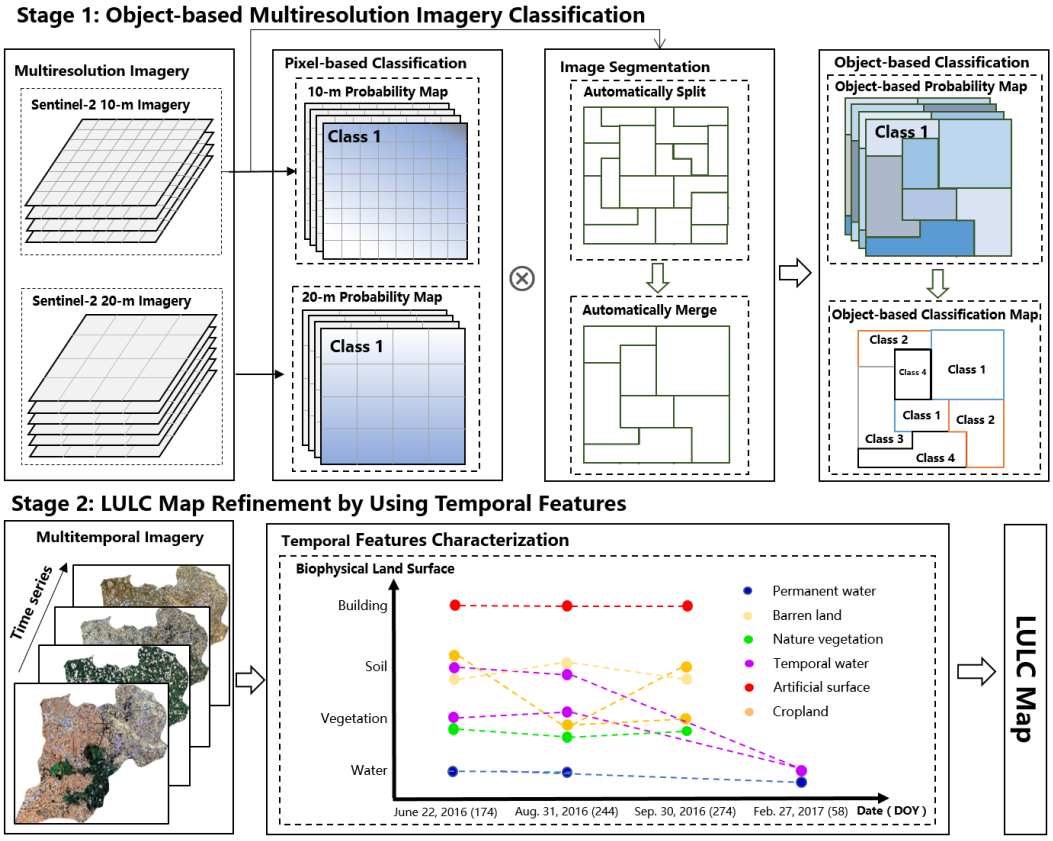


Fig. 2. Two-stage framework for LULC mapping using the multiresolution and multitemporal Sentinel-2 imagery.

training samples. In total, approximately 2000 pure pixels of per biophysical-based land-cover categories of water, vegetation, soil, and artificial surfaces are obtained from the Sentinel-2 10-m imagery, which correspond to approximately 500 pure pixels in the Sentinel-2 20-m imagery. In addition, to obtain a testing data set, we create a $10\times \sim 10$ grid for the study region and then randomly select 10 of the 100 subregions for truth map generation by digitalizing the land-cover boundaries based on the Sentinel-2 10-m imagery and Google Earth high-resolution imagery (see Fig. 1). After collection and preprocessing of the multiresolution and multitemporal Sentinel-2 images, a two-stage workflow for LULC mapping is carried out (see Fig. 2).

B. First Stage of Object-Based Multiresolution Imagery Classification

1) Autosegmentation Method for Higher Resolution Imagery: Image segmentation which splits an image into separated regions or objects depending on the specified parameters is a decisive procedure for subsequent object-based LULC classification [34], [43]–[45]. In this section, we conduct a split-merge strategy for automatic image segmentation using the Sentinel-2 10-m imagery.

First, we use the multiresolution segmentation (MRS) algorithm embedded into eCognition for initial segment generation [46], [47]. As a "bottom–up" image segmentation approach, the MRS algorithm begins with pixel-sized objects and is iteratively grown through the pairwise merging of the neighboring objects [27], [47]. The first step of MRS is the determination of three key parameters, namely, scale, color, and compactness. Scale is a threshold of heterogeneity allowed within the objects regarding color and shape properties. For each of the specified parameter values, the scale parameter varies greatly because it is the most influential parameter [48]. Generally, larger scale parameter values are associated with larger heterogeneity being allowed within objects, thus making larger and fewer objects [49]. The color and shape parameters range within the interval of 0–1 and are inversely proportional (i.e., color $= 1 -$ shape). The shape parameter depends on two other parameters, compactness and smoothness, which also range within the interval of 0–1 and are inversely proportional. In our study, the MRS method is used for a coarse segmentation, which is followed by automatic region merging for fine image segmentation. Specifically, the parameters of color and compactness are constantly set to 0.9 and 0.5, respectively, and the scale parameter is set very small, that is, 50, thus ensuring that all pixels within an image object belong to one land-cover class as often as possible.

The MRS algorithm is used to split the entire remote sensing images into small objects, thus facilitating the delineation of features of interest in the study area, such as the narrow channels of rivers and buildings with regular shapes. As some over-segmentation problems inevitably appear in this image-splitting procedure, an automatic region merge method is introduced (see Fig. 3). The existing algorithms usually require massive trial-and-error experiments for optimal parameter determination, which leads to a laborious region merging process. In this article, based on the training data

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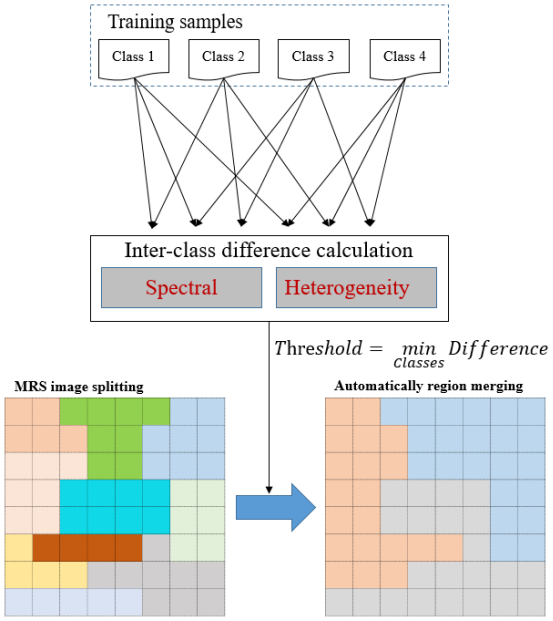


Fig. 3. Flowchart demonstrating the automatic procedure of region merging. The colors on the left and right represent the segments generated from MRS image-splitting and training data-based region merging procedures, respectively.

built for supervised object-based classification, we introduce an automatic method for neighboring segment merging by exploring whether the neighboring segments belong to the same land-cover class. First, we calculate the neighboring segment differences in terms of both spectral and spatial heterogeneity by making full use of the intersegment and intrasegment spectral angle information

$$\theta_{(S_1,S_2)} = \mathrm{SA}_{(\overline{S_1},\overline{S_2})} = \arccos \frac{\sum_{i=1}^{n} \left(\overline{S_1}\right)_i \left(\overline{S_2}\right)_i}{\sum_{i=1}^{n} \left(\overline{S_1}\right)_i^2 \sum_{i=1}^{n} \left(\overline{S_2}\right)_i^2} \quad (1)$$

$$H_{(S_1,S_2)} = \left| \frac{1}{M} \sqrt{\sum_{j=1}^{M} \left(\mathrm{SA}_{(S_{1j},\overline{S_1})}\right)^2} - \frac{1}{P} \sqrt{\sum_{h=1}^{P} \left(\mathrm{SA}_{(S_{2h},\overline{S_2})}\right)^2} \right| \quad (2)$$

where $\theta_{(S_1,S_2)}$ and $H_{(S_1,S_2)}$ are the metrics of the spectral difference and spatial heterogeneity difference, respectively; $\left(\overline{S_1}\right)_i$ and $\left(\overline{S_2}\right)_i$ represent the average spectral values of the segments S_1 and S_2 in band i, respectively; SA represents the spectral angle of two spectral vectors, which varies between 0 and $\pi/2$; S_{1_j} and S_{2_h} represent the spectral vectors within segments S_1 and S_2, respectively; and M and P represent the numbers of pixels contained in segments S_1 and S_2, respectively. Then, we calculated the minimum interclass differences according to (1)–(5) based on the training samples

$$T_{\mathrm{SA}} = \min\left(\mathrm{SA}_{(\mathrm{class}_a,\mathrm{class}_b)}\right) \quad (3)$$

$$T_H = \min\left(H_{(\mathrm{class}_a,\mathrm{class}_b)}\right) \quad (4)$$

where

$$a, b \in \left\{\mathrm{Tra}_{\mathrm{water}}, \mathrm{Tra}_{\mathrm{vegetation}}, \mathrm{Tra}_{\mathrm{soil}}, \mathrm{Tra}_{\mathrm{building}}\right\}, \quad a \neq b \quad (5)$$

and $\mathrm{SA}_{(\mathrm{class}_a,\mathrm{class}_b)}$ and $H_{(\mathrm{class}_a,\mathrm{class}_b)}$ represent the interclass spectral difference and the interclass heterogeneity difference, respectively. We define the minimum inter classes differences as the segment merging thresholds; thus, if the neighboring segment differences are less than the minimum interclasses differences, we can merge the neighboring objects. Particularly, to obtain relatively reliable merging thresholds, we divided the training samples into five portions that contain similar numbers of training samples of each land-cover type. According to (1)–(5), five pairs of interclass differences can be calculated, and these values are averaged to obtain the final merging parameters.

2) Probabilistic SVM Implementation for Multiresolution Imagery: As a typical supervised machine learning method, the SVM performs classification based on a hyperplane with a maximum margin, which is defined by a subset of points lying at the margins (called support vectors) [50]. Because the hyperplane maximizes the distance between different classes from a given set of training samples, it can accurately generalize unknown samples and minimize overfitting, which are the common problems when using classifiers, such as neural networks and decision trees [51], [52]. The SVM was initially proposed for binary classifications and to achieve multiclass classification applications, and a range of strategies, including one-against-all, one-against-one, and directed acyclic graphs, have been proposed [53]. To separate classes with nonlinear data distributions, kernel functions are used to transform training samples into a higher dimensional space in which linear class separation can be performed [54]. This also allows SVMs to perform well even when a limited amount and/or quality of training samples are available [55], [56]. In the SVM classification in our study, a multiclass one-against-one strategy with a radial basis function (RBF) kernel is applied. The kernel function requires two parameters, that is, C (penalty or regularization term) and γ (controls the width of the kernel function), which control the flexibility of the classifier. In this article, the optimal parameters are determined by a grid search in conjunction with an internal threefold cross-validation using the average overall accuracy (OA) as a measure of performance. The parameter space is discretized along 2^d, with $d = 0, 0.5, 1, \ldots, 8$ for C and $d = -4, -3.5, \ldots, +4$ for γ. Nevertheless, unlike the common SVM method, which predicts only the class label as the final classification result, an extended SVM method based on combining all pairwise comparisons is used for class posterior probability estimation in our study [57], [58]. Through probabilistic SVM implementation using the open-source LIBSVM library [59], probability maps for land-cover classes are produced based on the Sentinel-2 10-m imagery and 20-m imagery, respectively.

Object-based

3) Area-Weighted Method for Per-Object Classification: classification begins with image segmentation to aggregate pixels into larger image objects, and the objects are then categorized according to the spectral and spatial features in the imagery. Based on the generated objects and multiresolution class probability maps, in this section, we present a novel object-based analysis scheme to determine the land-cover category of each object (see Fig. 2). By contrasting the Sentinel-2 10- and 20-m imageries, the 20-m imagery, which possesses

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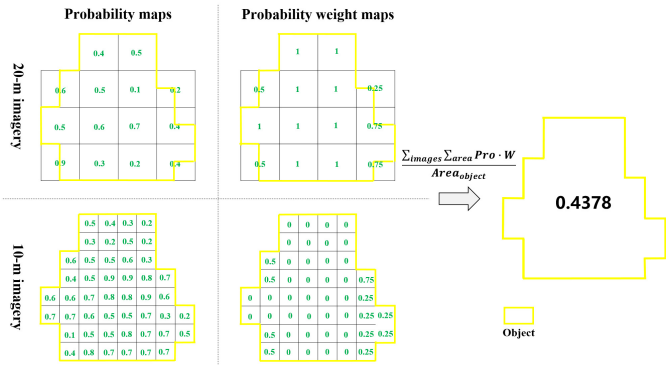


Fig. 4. Object-based analysis performed by integrating the Sentinel-2 10- and 20-m probabilistic maps.

abundant spectral information, can produce more reliable SVM posterior probabilities, and the finer resolution 10-m imagery, which has more detailed spatial information, can facilitate finer recognition of land-cover boundaries. As mentioned in Section III-B1), we conduct image segmentation using the Sentinel-2 10-m imagery, and it is likely that the 20-m probability map will be fragmented by the different segmentation objects. To combine the properties of the 10- and 20-m imageries, an area-weighted average method is developed to calculate the per-object class probability in our study (see Fig. 4). Specifically, the probability weights of the 20-m imagery pixels are determined by the per-pixel area of the 20-m pixel to be contained in the segmentation object, and the weights of the 10-m imagery pixels are obtained through an inversely proportional calculation [see (6) and Fig. 4]. In addition, an area-weighted method based on the object probabilities derived from the respective 10-m probability map and 20-m probability map is carried out. The determination of probability weights for the 10-m imagery and 20-m imagery and the following area-weighted method are shown in Fig. 4 and in the following equations:

$$w_{20\ \mathrm{m}} = \frac{\mathrm{area}_{\mathrm{encloed}}}{\mathrm{area}_{20\ \mathrm{m}}},\ w_{10\ \mathrm{m}} = 1 - w_{20\ \mathrm{m}} \tag{6}$$

$$\mathrm{Pro}_{\mathrm{object}} = \frac{\sum_{i=1}^{n} \mathrm{Pro}_{20\,\mathrm{m}} \cdot w_{20\,\mathrm{m}} \cdot \mathrm{area}_{\mathrm{encloed}} + \sum_{i=1}^{m} \mathrm{Pro}_{10\,\mathrm{m}} \cdot w_{10\,\mathrm{m}} \cdot \mathrm{area}_{10\,\mathrm{m}}}{\mathrm{Area}_{\mathrm{object}}} \tag{7}$$

where $\mathrm{area}_{\mathrm{encloed}}$ represents the 20-m pixel area to be enclosed in the object; $\mathrm{area}_{20\ \mathrm{m}}$ and $\mathrm{area}_{10\ \mathrm{m}}$ represent the pixel areas of the 20-m imagery and 10-m imagery, respectively; n and m represent the numbers of pixels contained in the objects of the 10- and 20-m probability maps, respectively;$\mathrm{Pro}_{10\ \mathrm{m}}$ and $\mathrm{Pro}_{20\ \mathrm{m}}$ are the SVM class probabilities derived from the Sentinel-2 10- and 20-m imageries, respectively; and $w_{10\ \mathrm{m}}$ and $w_{20\ \mathrm{m}}$ denote the corresponding probability weights, respectively.

C. Second Stage of LULC Map Refinement Using Temporal Features

Four biophysical-related land-cover classes of water, vegetation, soil, and artificial surface can be obtained through the implementation of multiresolution imagery classification. These four land-cover types can be regarded as the base classes because almost all LULC categories (e.g., crop, forest, and lake) can be affiliated with them. Prior to setting up the Xiongan New Area, the LULC in this region was dominated by cropland, that is, land growing mainly corn and wheat. Based on the land investigation materials provided by the local government, the seeding stage of corn is closely followed by the harvesting time of wheat, which occurs approximately in June. With about three months of growing, the corn becomes mature, and harvesting follows. During the busy season, continual human activities have been imposed on the agricultural land and lead to a fast transition between soil and vegetation. In addition, as the largest freshwater wetland in North China, Baiyangdian forms a crucial ecosystem for water conservation and provides a significant habitat benefitting wetland biodiversity conservation. Baiyangdian is a seasonally inundated wetland; hence, the range of water storage is associated with the land surface conditions, human activities (e.g., overgrazing, wetland drainage), and climate change [60]. Based on the materials produced in recent years, although the annual rainfall is concentrated in summer, the sustained high temperatures and massive agricultural water needs of this region make water shortage frequently occur in summer. Therefore, in this article, winter is defined as the wet season because it is characterized by more surface water coverage and summer is defined as the dry season. Based on the above information, we adopt multitemporal Sentinel-2 images to represent specific periods of time for the study region, and then the temporal features corresponding to land use categories can be summarized, as in Table II. In addition, the fine LULC categories can be determined by the following algorithm:

$$\begin{cases} \text{Permanent water}, \mathrm{Class}_{58,174,244} = \text{Water} \\ \text{Temporal water}, \ \mathrm{Class}_{58} = \text{Water} \\ \qquad \cap \mathrm{Class}_{174,244} \in \text{Vegetation, Soil} \\ \text{Natural vegetation}, \mathrm{Class}_{174,244,274} = \text{Vegetation} \\ \text{Barren land}, \mathrm{Class}_{174,244,274} = \text{Soil} \\ \text{Building}, \mathrm{Class}_{174,244,274} = \text{Artificial surface} \\ \text{Cropland}, \left(\mathrm{Class}_{174} \neq \hat{\mathrm{A}}, \mathrm{Class}_{244}\right) \\ \qquad \cup \left(\mathrm{Class}_{244} \neq \hat{\mathrm{A}}, \mathrm{Class}_{274}\right) \\ \quad \cap \left(\mathrm{Class}_{174,244,27} \in \text{Vegetation, Soil}\right) \end{cases}$$

where the term class represents the biophysical land surface states in the range of {water, vegetation, soil, and artificial surface} and the subscripts represent the day of the year (DOY) corresponding to the multitemporal images. Through a combination of temporal features, a refined LULC map can finally be produced.

D. Accuracy Assessment

Ten randomly distributed regions of the study area are selected for reference map generation (see Fig. 1). By referring to Google Earth high-resolution imagery and auxiliary materials, the ground-truth map is obtained through manually digitizing the land-cover boundary on-screen. The obtained ground-truth map is sampled to a raster image with a spatial resolution of 10 m, and pixels at land-cover boundaries are

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TABLE II
RULESET FOR TEMPORAL FEATURES' EXTRACTION

LULC classes	Temporal features	Acquisition dates of multitemporal images (DOY)	Ruleset LULC classification scheme
Permanent water	Maintain stability in both the dry season and wet season	174 244 58	Classified as surface water in all multitemporal imagery
Temporal water	Water surface turns into the land surface from the dry season to wet season		Classified as only water in the imagery acquired on Feb. 27, 2017
Natural vegetation	Maintain stability in the growing season	174 244 274	Classified as vegetation in all multitemporal imagery
Barren land	Maintain stability in the long term		Classified as soil in all multitemporal imagery
Building	Maintain stability in the long term		Classified as buildings in all multitemporal imagery
Cropland	Rapid transition between soil and vegetation in harvesting season		Classified as land surface changes between vegetation and soil during the planting and harvest season

assigned to the land-cover category that accounts for the largest area of the pixel.

For comparison, two groups of the LULC classification are conducted in our study: 1) single-resolution imagery-based classification and 2) multiresolution-imagery-based LULC classification. We use two classification strategies for the LULC classification: 1) the pixel-based and 2) objected-based strategies. Particularly, the MRS, which is embedded in eCognition, is used for object generation [46], [47]. To perform comparisons with the existing object-based LULC classification methods, the general object-based and the MV objected-based methods are compared. In each LULC classification group, three classical classifiers, that is, SVM, random forest (RF), and k-nearest neighbor (KNN), are implemented using the Python scikit-learn package [61] in our study. To avoid any spatial information loss in the LULC classification, the Sentinel-2 multiresolution imagery is unified through upsampling of the 20-m imagery to 10-m imagery. For quantitative assessment of the newly proposed method, two overall indicators, that is, OA and average accuracy (AA), are adopted in our study. Moreover, to examine whether the new method significantly improves the accuracy compared with other LULC mapping methods, a McNemar's statistical test is carried out in our study. Specifically, McNemar's continuity corrected chi-square statistic is computed as shown in the following equation [62], [63]:

$$\chi^2 = \frac{(|f_{12} - f_{21}| - 1)^2}{f_{12} + f_{21}} \tag{8}$$

where f_{12} and f_{21} denote the frequencies of cases that are correctly classified by one classification method but wrongly classified by the other.

TABLE III
SUMMARY OF MCNEMAR'S CONTINUITY-CORRECTED CHI-SQUARE STATISTIC TEST FOR DIFFERENCES IN CLASSIFICATION ACCURACY

LULC mapping	Single-resolution imagery-based		Multiresolution imagery-based		
	10-m KNN	20-m SVM	SVM	MRS+KNN	MRS+SVM+MV
New method	χ^2				
	60601	29858	4903	40461	18476
	P-value				
	0	0	0	0	0

IV. RESULTS

A. Comparisons of Multiresolution Imagery Classification and Existing Land-Cover Mapping Methods

Through the implementation of the proposed multiresolution imagery classification, four biophysical-related land-cover categories, including water, vegetation, soil, and artificial surfaces, can be classified using the multiresolution Sentinel-2 imagery. Particularly, we select the land-cover map corresponding to the date of September 30, 2016, for accuracy analysis in our study. As the minimum unit of an image lager, mixture problem becomes more serious; to minimize the impact caused by mixed pixels in the accuracy assessment, all classification maps are resampled to a 10-m spatial resolution to maintain consistency with the reference map. According to the quantitatively accuracy assessment, the proposed multiresolution imagery classification achieved the highest classification accuracy, with an OA of 88.58% and an AA of 87.88%, and it is followed by the multiresolution-imagery-based SVM classification (OA 86.77% and AA 85.72%) and multiresolution-imagery-based MRS + MV SVM classification (OA 85.02% and AA 83.49%). McNemar's chi-square test, which accounts for pairwise classification comparisons, is included in Table III, further demonstrating that the newly proposed method statistically outperforms the 10-m imagery-based KNN, 20-m imagery-based SVM, multiresolution-imagery-based SVM, multiresolution-imagery-based MRS + KNN, and multiresolution-imagery-based MRS + SVM + MV methods, and the P-value for comparisons with each method is less than 0.05. More details about the accuracy assessment are presented in Table IV.

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TABLE IV
CLASSIFICATION ACCURACIES DERIVED USING THE NEWLY PROPOSED METHOD AND OTHER COMMONLY USED METHODS

Class/ Accuracy (%)	Single-resolution imagery-based						Multiresolution imagery-based									New method
	10-m imagery			20-m imagery			SVM	RF	KNN	MRS			MRS+MV			
	SVM	RF	KNN	SVM	RF	KNN				SVM	RF	KNN	SVM	RF	KNN	
Water	95.36	76.24	74.91	85.35	89.64	90.94	85.35	73.99	74.92	89.80	82.40	87.70	80.09	70.22	72.86	92.15
Vegetation	96.76	89.14	90.23	91.09	75.49	91.82	92.24	91.24	90.22	86.11	88.63	89.79	90.16	88.31	89.24	94.15
Soil	77.77	68.28	74.31	75.17	83.84	81.62	84.07	63.29	74.31	74.97	70.12	77.68	82.34	63.18	75.70	83.61
Artificial surface	35.82	82.40	79.23	85.08	72.30	83.34	81.21	83.21	79.23	85.22	71.47	77.57	81.36	84.74	83.00	81.56
AA	76.43	79.02	79.67	84.17	80.32	86.93	85.72	77.93	79.67	84.02	78.15	83.18	83.49	76.61	80.20	**87.88**
OA	80.35	80.01	81.91	84.09	78.13	83.38	86.77	79.52	81.91	82.00	78.94	83.15	85.02	78.21	82.41	**88.58**

The multiresolution-imagery-based SVM classification, which achieves the best performance among the comparison methods, is selected for qualitative comparative analysis in this section. Because both the Sentinel-2 10-m imagery (four bands) and Sentinel-2 20-m imagery (nine bands) are commonly used for land-cover classification practice, the 10-m Sentinel-2-imagery-based classification and 20-m Sentinel-2-imagery-based SVM classification are also selected for qualitative analysis in this section. Specifically, the 9-band 20-m imagery is acquired by layer stacking with a unified 20-m pixel size, and the repetitive near-infrared band of the native 20-m resolution is removed. To unify the pixel size of the multiresolution imagery, the 20-m imagery is upsampled to the 10-m imagery. Through the implementation of the proposed multiresolution imagery classification method, we can obtain an overview land-cover map of the Xiongan New Area [see Fig. 5(a)]. To illustrate the mapping quality of the new method, three local regions are selected for comparative analysis in our study. The results show that the classification map derived from the 10-m imagery suffers from mass confusion between soil and artificial surfaces [see Fig. 5(f) and (v)] and between water and vegetation [see Fig. 5(m)]. According to Fig. 5(g), (o), and (v), the 20-m imagery-based classification achieves relatively high accuracy. However, the map shows a deficiency in subtle surface feature recognition. For example, there are mapping omissions of slender rivers and vegetation [see Fig. 5(n) and (u)]. The classification maps derived from the combination of the 10- and 20-m imageries show unstable accuracy for different landscape scenes [see Fig. 5(h), (o), and (v)]. Compared with the classification map derived from the 20-m imagery, the combined images show a better classification result in test scenario 1 [see Fig. 5(h)] and a worse result in test scenarios 2 and 3 [see Fig. 5(o) and (v)]. From visual inspection of the classification maps, the land-cover map derived using the proposed method exhibits the best performance because it inherits the properties of high-accuracy classification and subtle feature recognition using the Sentinel-2 20-m imagery and 10-m imagery [see Fig. 5(e), (l), and (s)].

It is worth noting that the image segmentation contributes to the final classification results as shown in Fig. 5(c), (d), (j), (k), (q), and (r), and the proposed automatic image segmentation obtained the desired results with few under-segmentation errors. Specifically, through automatic image segmentation, the easily confused land-cover categories of artificial surface and soil [see Fig. 5(d) and (r)] as well as those of surface water and vegetation [see Fig. 5(k)] can be identified accurately. The proposed automatic image segmentation also achieved acceptable results in suppressing over-segmentation; that is, scattered and isolated pixels can be eliminated through image segmentation, thereby avoiding much of the salt-pepper noise produced in pixel-wise land-cover classification.

B. Results of LULC Map Refinement by Combining Temporal Features in the Xiongan New Area

The LULC mapping in Xiongan New Area demonstrates that the temporal features are helpful for LULC map enhancement, specifically by applying the temporal features corresponding to the land use categories, a refined LULC map that contains the six subdivided land covers of permanent water, temporal water, natural vegetation, barren land, building, and cropland (see Fig. 6) can be produced. To verify the effectiveness of the refined LULC map, extensive inspection of the refined image classification result is conducted based on the multitemporal Seninel-2 images, high-resolution Google Earth image, and land investigation materials provided by the local government. According to the statistics, the Xiongan New Area is dominated by cropland, which has a total area of up to 51.59%. In addition, the Xiongan New Area shows a well-protected eco-environment with a high ecological land coverage proportion of 31.93% (including natural vegetation coverage of 22.38%, temporal water coverage of 6.22%, and permanent water coverage of 3.33%). According to the refined

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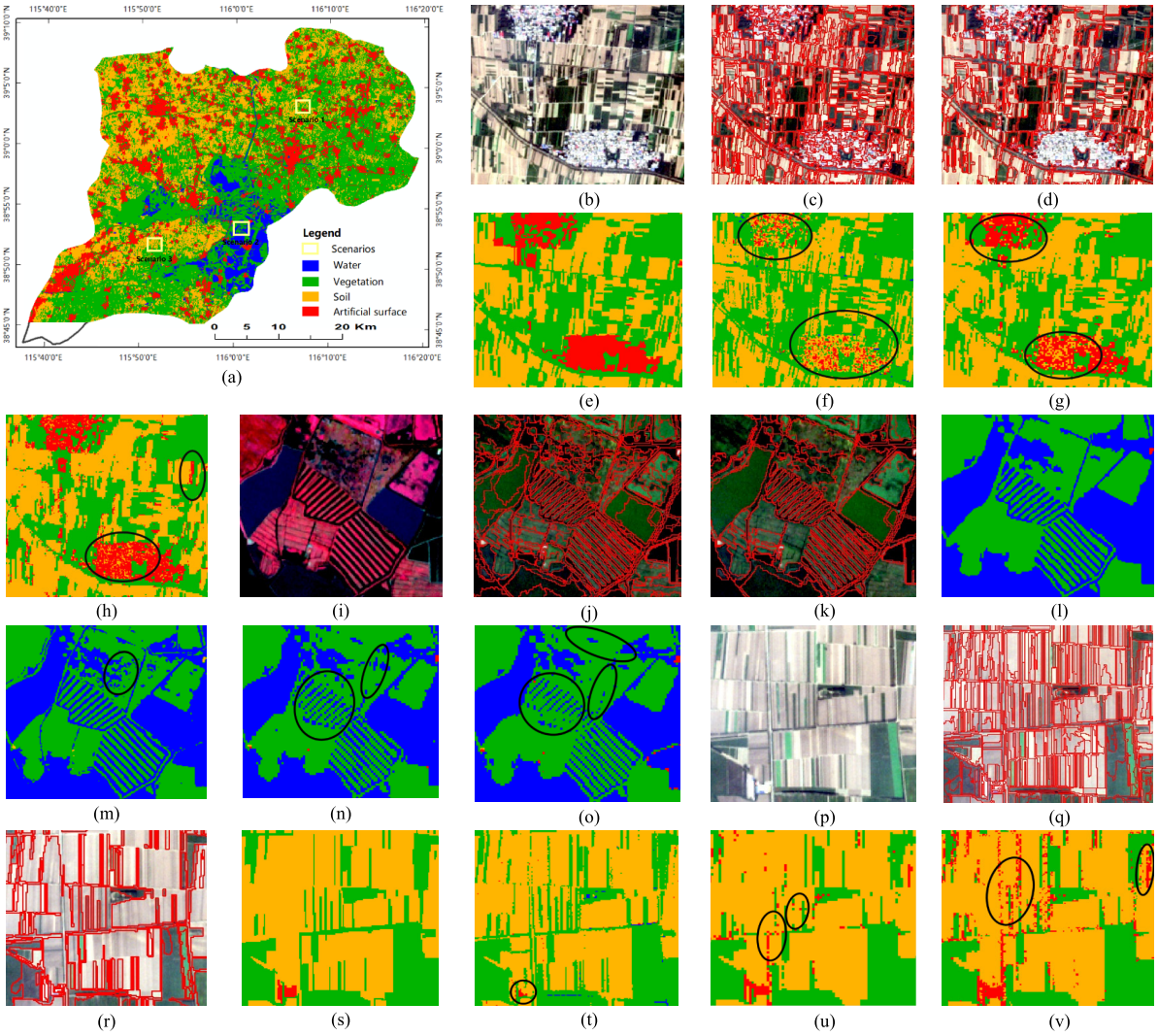


Fig. 5. Land-cover classification map and three scenarios for the method comparisons. (a) Land-cover map derived by multiresolution imagery classification. (b) Scenario 1. (i) Imagery is displayed as a false-color composite (RGB: bands 8a, 4, 3). (c), (j), and (q) Segmentation results through automatic image splitting. (d), (k), and (r) Segmentation results through automatic region merging. (e), (l), and (s) Classification result derived from the proposed method. (f), (m), and (t) SVM-based classification result with the Sentinel-2 10-m imagery. (g), (n), and (u) SVM-based classification result with the Sentinel-2 20-m imagery. (h), (o), and (v) SVM-based classification result with the combined Sentinel-2 10- and 20-m imageries. (p) Scenario 3.

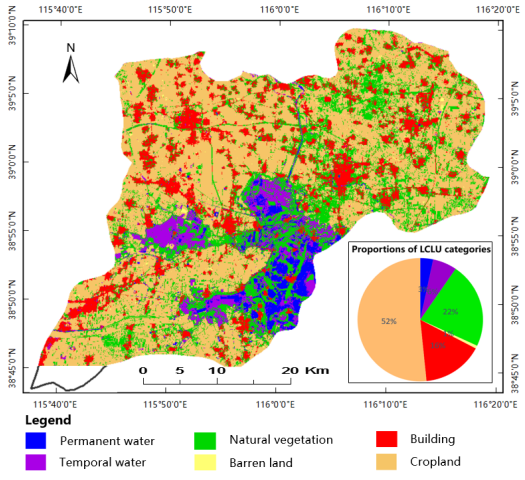


Fig. 6. Refined LULC map for the Xiongan New Area.

LULC map, the Xiongan New Area is still less developed because the 15% building area is spatially dispersed.

V. Discussion

A. Does Atmospheric Correction Affect LULC Mapping?

According to the official document [42], the atmospheric-corrected (including adjustments for thin cirrus, slope, and adjacency effects) Sentinel-2 imagery achieved a significant improvement in image quality compared with the nonatmospheric-corrected imagery, particularly in cloudy regions. To evaluate the effect of atmospheric correction on the proposed LULC mapping method, a comparison of the Sentinel-2 TOA reflectance imagery and the atmospheric-corrected Sentinel-2 BOA reflectance imagery is performed. The SVM-based probabilistic estimation and automatic image segmentation are conduct using TOA reflectance imagery and BOA reflectance imagery as shown in Fig. 7. Except for a slight difference in the building probability maps [see Fig. 7(a4) and (b4)], the differences between the other pairwise estimated probability maps are small. Moreover, as shown in Fig. 7(a6) and (b6), the differences

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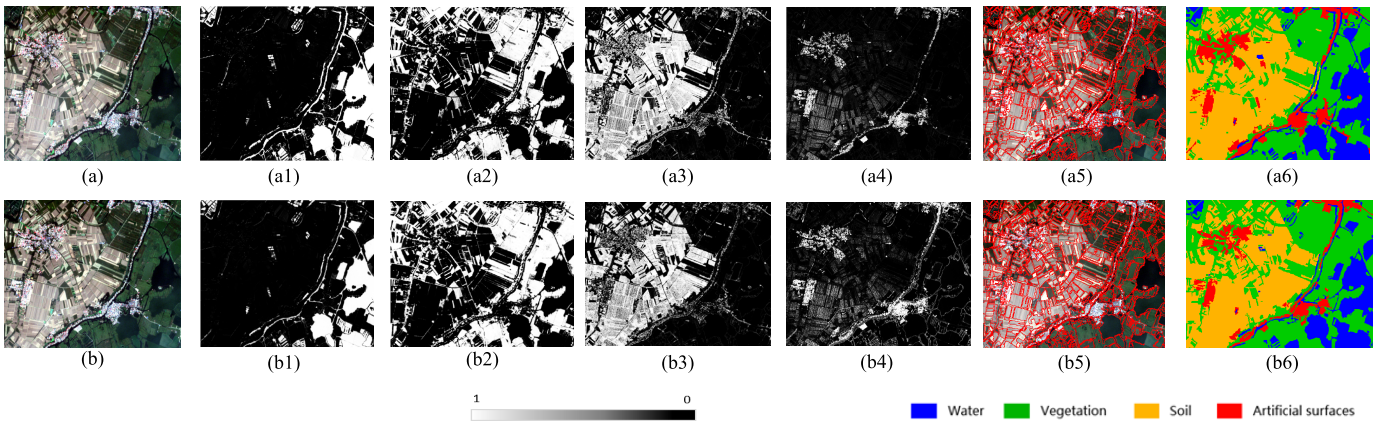


Fig. 7. Results derived from SVM-based probabilistic estimation and image segmentation using (a) TOA reflectance image and (b) BOA reflectance image, respectively. (a1) and (b1) Water probability map. (a2) and (b2) Vegetation probability map. (a3) and (b3) Soil probability map. (a4) and (b4) Building probability map. (a5) and (b5) Image segmentation. (a6) and (b6) Object-based classification.

between the image segmentation results using the Sentinel-2 TOA and the BOA reflectance imageries are also not significant, with 390 and 379 segments generated through the image segmentation, respectively. Based on the SVM-based probabilistic estimation and automatic image segmentation, the object-oriented classifications are implemented using the Sentinel-2 TOA and BOA reflectance imageries, respectively. As shown in Fig. 7(a6) and (a7), a slight difference can be observed between the classifications maps, and the quantitative accuracy assessment showed that the accuracy (87.68%) of the classification result obtained based on the BOA reflectance image is slightly higher than the accuracy (86.58%) obtained using the TOA reflectance imagery. In summary, although there is no significant difference between the LULC mapping results based on the atmospheric-corrected and nonatmospheric-corrected Sentinel-2 imagery, the LULC mapping is inevitably affected by the atmospheric correction procedure. The atmospheric-corrected image-based classification achieves a slightly better result than the nonatmospheric-corrected image in our study, and therefore, before full evidence about how the atmospheric correction affects the Sentinel-2 imagery classification is implemented, we recommend that atmospheric correction should be undertaken when applying the Sentinel-2 imagery to the LULC classification.

B. Extendibility of the Proposed LULC Mapping Framework

In this article, a new LULC classification framework that consists of image segmentation, multiresolution imagery classification, object-based area weighting, and temporal features characterization is proposed. The proposed framework is highly extensible, and with the development of related techniques, many advanced algorithms can be easily integrated into the proposed framework, thus providing the potential to improve its capability in LULC mapping. In the current version of the proposed framework, the image segmentation and class probability estimation are performed using an automatic split-merge strategy and SVM algorithm, respectively. However, to extend the capability of the proposed framework, these algorithms can be easily replaced by other advanced algorithms. We use the pixel area of multiresolution imagery contained in the object to determine the probability weights, followed by an area-weighting method for object posterior probability estimation. In addition to the area-weighting method, the spectral bands of each multiresolution image also affect the optimal probability weight determination, and thus the related determination rule for probability weight can be further explored to improve the performance of LULC mapping using the proposed framework. We make a trade-off based on the influence analysis of image segmentation because image under-segmentation will inevitably lead to misclassification of the image while over-segmentation has a limited influence on the final classification result if the image classification method is appropriate. To prevent image classification from being affected by image segmentation, our strategy is to strongly prevent image under-segmentation and therefore allow a degree of image over-segmentation. Therefore, the automatic image segmentation method is worthy of further study to promote a better balance between image under-segmentation and over-segmentation. Moreover, although the proposed framework was initially designed for the multiresolution and multitemporal Sentinel-2 imagery, the proposed framework is also highly extensible in terms of data applicability. As multisensor images become more exploitable, the proposed framework can be directly extended to multisensory and multiresolution image classification on the premise of high-precision image-to-image registration.

VI. Conclusion

In this article, we used a novel and efficient LULC classification method using multiresolution and multitemporal Sentinel-2 images. The new approach fully exploits the spectral, spatial, and temporal information of the Sentinel-2 imagery through: 1) automatic image segmentation based on the Sentinel-2 10-m imagery; 2) multiclass posterior probability estimation based on Sentinel-2 multiresolution; 3) object-based LULC mapping based on multiresolution probability maps; and 4) temporal feature extraction for refined LULC classification based on the Sentinel-2 multitemporal imagery. Three conventional LULC classification algorithms are compared with the new approach, and the proposed method achieved the highest OA of 88.58% and an AA of 87.88%.

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We applied the proposed method to map the newly established special economic zone of the Xiongan New Area, and the resulting LULC map demonstrates that the Xiongan New Area is currently in a "less developed" state and has a preserved ecosystem, particularly in the region surrounding the Baiyangdian wetland. Based on the area statistics of the LULC categories, ecologically beneficial land coverage includes natural vegetation, temporal water, and permanent water, which account for 22.38%, 6.22%, and 3.33% of the total land area, respectively. Only 15.69% of the land surface is developed into built-up coverage in the current Xiongan New Area. As mentioned above, the LULC map and the related statistical results can also be regarded as base data to assist with urban planning and ecological environmental monitoring. With the forthcoming rapid development of the Xiongan New Area, our future work will focus on LULC classification improvement and dynamic monitoring of the LULC changes in the Xiongan New Area based on the framework proposed in this article.

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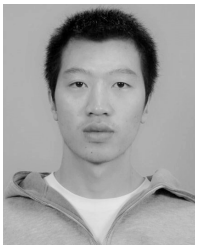
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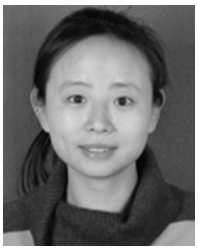


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